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An Alzheimer's disease classification model using transfer learning Densenet with embedded healthcare decision support system

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ABSTRACT

Training a Convolutional Neural Network (CNN) from scratch is time-consuming and expensive. In this study, we propose implementing the DenseNet architecture for classification of AD in three classes. Our approach leverages transfer learning architectures as the base model and showcases superior performance on the MRI dataset compared to other techniques. We use a variety of methodologies to provide a thorough study of our model. We first create a performance baseline without data augmentation, addressing the difficulties in classifying Alzheimer's disease (AD) caused by high-dimensional MRI brain scans. The improved model performance obtained through data augmentation is then highlighted, demonstrating its effectiveness in handling sparse data and assisting in generalization. We also investigate the impact of omitting particular transformations and modifying dataset split ratios, providing more insights into the behavior of the model. Through comprehensive evaluation, we demonstrate that our proposed system model achieves an accuracy of 96.5% and an impressive AUC of 99%, surpassing previous methods. This study mainly highlights the effectiveness of DenseNet architecture, current study limitations and future recommendation. Moreover, incorporating a healthcare decision support system further aid in providing valuable insights for AD diagnosis and decision-making in clinical settings.

1. Introduction

Millions of people throughout the world are afflicted by the common neurodegenerative ailment known as Alzheimer's disease [1]. Physicians typically monitor AD using the Clinical Dementia Rating (CDR) system based on signs and symptoms. Subjects are divided into three states using the CDR, Vemuri et al. [2] including CN (Cognitive Normal), MCI (Mild-Cognitive Impairment), and AD. Atrophy of the GM(Gray Matter) and WM (White Matter) and an increase in CSF (Cerebrospinal Fluid) volume are markers of AD progression. The cerebral cortex and hippocampus get smaller as the size of the ventricles in the brain grows in AD patients. Epistolaryngeal memory and spatial memory are harmed when the hippocampus's size is decreased. Planning, judgement, and short-term memory impairments are the result of this neuronal injury [3]. This decrease leads to dysfunctional synapses, damaged terminals, and further cell death.

Accurate early AD diagnosis is essential for prompting appropriate therapies and improving patient outcomes [4]. The development of machine learning and artificial intelligence has made decision support systems into potent instruments in clinical settings. These systems use AI algorithms to assess patient data, including clinical data and medical image data, and then deliver insightful recommendations to healthcare practitioners [5,6]. In order to identify illnesses and generate precise forecasts, machine learning systems examine patient data, including medical images [7–9]. This promotes early identification [5], individualized care, and better patient outcomes. Machine learning algorithms, for instance, may be trained on annotated pictures used in medical imaging to automatically detect and categorize diseases including cancer, heart issues, and neurological disorders [10]. Healthcare professionals can improve patient care by integrating AI-based decision support systems into clinical practice. These technologies can help diagnose patients more accurately, optimize treatment regimens, and optimize costs [11]. Additionally, healthcare decision support systems

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are essential for assisting medical practitioners in difficult decision-making procedures [12]. These systems offer practitioners individualized treatment suggestions and therapeutic alternatives by integrating clinical standards, evidence-based medicine, and real-time patient data [13].

One specific area of focus within healthcare decision support systems is AI-based decision support for diagnosis [14] such as leveraging machine learning algorithms where it has helped risk assessment, illness diagnosis, and categorizing patients as having a condition or not [15]. However, Deep Learning [16] on the other hand is subset of ML, presents even greater potential due to its capability to automatically learn hierarchical representations from even more complex data [17]. For instance, CNNs [18], a type of DL model, have achieved great success in medical image processing tasks, allowing for precise classification of disease. Deep learning offers considerable potential for boosting diagnostics and enhancing healthcare outcomes due to its greater capacity to extract complex patterns and characteristics from medical pictures. This ability is especially important in the realm of medical imaging [17], where high-dimensional data, such as MRI [19] or CT scans [20], calls for sophisticated processing methods. By enhancing diagnostic precision, streamlining treatment plans, and lowering healthcare costs, the use of AI-based decision support systems in clinical settings has the potential to revolutionize healthcare [21].

Another key element that further improves the capabilities of DL-based models is known as Transfer Learning (TL) [22]. Transfer learning enables the use of knowledge and representations acquired from one task or dataset on another task that is relevant. The use of pre-trained CNN models in medical imaging is made possible by transfer learning and large-scale [23] datasets like ImageNet. By leveraging the learned features and representations from these pre-trained models, healthcare practitioners can enhance the performance of their own diagnostic models, even with limited labeled medical imaging data. A pre-trained model is used as the foundation for a new model. For instance, a pre-trained model [24] is trained with large amounts of data like ImageNet [25] which consists of millions of images with more than 1000 labels. And its desirable and a good choice to transfer the domain knowledge from these models to another domain like classification task in medical area as a feature extractor of general features [26]. Since we are dealing with medical types of data its and the pre-trained models usually on different data like ImageNet, we cannot fully use the pre-trained model instead we need to add another layer of CNN as domain-specific feature extractor and modify the output layer as per domain-specific problem [27]. When compared to training with randomly initialized models, knowledge transfer requires fewer training instances to complete a different task, takes less time, and improves accuracy. Various state-of-the-art architectures such as Dense Net [28], AlexNet [29], Inception [30], VGG [31], ResNet [32] which can be fine-tuned to solve other problems with a much smaller dataset and in much less time. In our specific research, the focus lies on leveraging the capabilities of DenseNet, a prominent deep learning architecture, to develop an Alzheimer's disease classification.

This study introduces a method for classifying three stages of Alzheimer's Disease, DenseNet, a powerful DL-based architecture which can empower the decision support systems in clinical settings. By fine-tuning a DenseNet model, our goal is to extract and utilize relevant features from MR images, enabling effective differentiation between AD, NC, and MCI cases. The suggested approach has a number of benefits over conventional AD classification techniques. Firstly, it decreases the dependence on extensive labeled data, which is often scarce in medical domains. Furthermore, it makes use of the learned representations from a large-scale dataset, allowing the model to detect subtle anomalies and detailed patterns in medical images. Finally, it offers a comprehensive and precise categorization structure to support doctors in arriving at fast and correct diagnoses. This work demonstrates the use and effectiveness of our proposed model using the dataset we collected from an online repository comprising MRI scans from AD, MCI patients and healthy individuals. We evaluate the model's performance in terms of accuracy, sensitivity, specificity, and other relevant metrics, comparing it to existing approaches in the field. By contrasting it with current methods in the area, we assess the model's performance in terms of accuracy, sensitivity, specificity, and other pertinent metrics.

To thoroughly evaluate the model's effectiveness, we present four different evaluation methods. Starting with evaluation of models without applying any data augmentation technique, evaluation of Models with the use of Data Augmentation, Analysis on the basis of selective augmentation, and how the ratio of training to testing affects the performance. Our results demonstrate the potential of DenseNet-based transfer learning, offering a significant improvement in AD classification. The implications of this strategy go beyond AD diagnosis and treatment, promising improved patient care and administration. This study highlights the transformative potential of DenseNet-based transfer learning and contributes fundamentally to the field of AD classification.

The remainder of the paper is structured as follows. In Section 2, we highlight the motivation and significance of employing deep learning in AD classification. Section 3 presents the contributions of our work, emphasizing the use of DenseNet for image-based analysis and the significance of transfer learning. Section 4 provides a concise overview of related work in AD diagnosis, setting the context for our study. In Section 5, we detail the materials and methods used in our experiments. The results and analysis of these experiments are presented in Section 6. In Section 7, which is the last section, we provide the implications of our findings and provide concluding remarks.

2. Motivation

Alzheimer's disease poses a major challenge for clinicians due to its complex nature and the necessity for early identification. Traditional approaches and human intuition often fall short in accurately predicting and visualizing the progression of the disease. To overcome this issue, computationally demanding and unconventional approaches like deep learning are needed. Medical experts are increasingly relying on DL techniques for usage in both disease prediction and visualization in order to deliver foresighted and tailored treatment regimens [33]. This shift helps doctors decide on treatments and health economists analyze costs, both of which improve patients' lives.

Healthcare decision support systems in clinical settings play a vital role in leveraging deep learning techniques for accurate and effective models in AD diagnosis. Deep have emerged as leading machine learning approaches for visual object detection. While CNNs were introduced over two decades ago, despite the fact that they were first presented over 20 years ago, recent advancements in hardware, software, and network structures have enabled the training of truly deep CNNs [34]. Only lately have advancements in computer hardware, software, and algorithmic network structure made it possible to train truly deep CNNs [35]. However, as CNNs grows deeper, they encounter challenges such as information vanishing or loss of gradients along the network path [36], Addressing these challenges becomes crucial for developing accurate and effective models for Alzheimer's disease diagnosis. According to Andrew Ng in 2016 conference about [37] "Nuts and Bolts of Building AI Applications using Deep Learning" Transfer learning will succeed supervised learning as the next major factor in AI-based machine and DL commercial success. Conventional ML and DL algorithms were only designed to solve separate specific tasks, which require a lot of data. Transfer learning was developed to overcome the isolated learning paradigm. A study in 2017 [38] conducted by Lopez and Valiati, demonstrated the use of pre-trained CNNs as feature extractors for detection of tuberculosis which can produce reasonable results without a large dataset, an expensive, time-consuming training phase, or both.

In the context of AD classification, our motivation stems from the need to improve prediction accuracy and address the challenges

posed by limited training data. By exploring the application of transfer learning techniques, specifically the DenseNet architecture, we aim to leverage the learned features from pre-trained models and enhance the classification performance using MRI scans. We also extend our investigation to explore three different variants of the DenseNet architecture, evaluating their respective performances in AD classification. Our goal is to develop a robust and efficient AD classification model by DenseNet model and addressing the limitations of conventional approaches. By harnessing the power of deep learning and transfer learning, we strive to improve the accuracy of AD diagnosis, enable early identification, and enhance the decision-making process for clinicians.

3. Contribution

People with Alzheimer's disease are identified in our research work, and our goal is to identify people who may have the disease early. The dataset for AD is available on Kaggle which is used for training all patient's data using DenseNet architectures, to identify the affected people quickly and efficiently in a highly effective manner. We then examine another transfer learning paradigm to capture generic and domain-specific features and reserve using pre-trained DenseNet on ImageNet and learn new specialized representations by replacing top layers according to our output classifier. The dataset's tiny size is the biggest problem. The main obstacle that must be overcome is the amount of dataset which is small. We used data augmentation technique to help the model's function at their optimal level and dramatically enhance their effectiveness during classification. The paper can be used to infer the following contributions:

- Present an overview and significance of DenseNet architecture for image-based analysis.
- Implement three versions of DenseNet architecture, including DenseNet-121, DenseNet169, and DenseNet-201.
- We present four different evaluation methods to evaluate the model's effectiveness.
- On-fly data augmentation is used to avoid the overfitting problem.
- Incorporate a healthcare decision support system for diagnosis and decision-making in clinical settings.

4. Transfer learning

In the context of training of DL-based models, transfer learning is a powerful method that makes use of information acquired from one domain to enhance performance on a separate but related task or domain. It is especially helpful when obtaining labeled data is difficult or expensive. TL facilitates the creation of highly accurate and effective models with shorter training times by transferring acquired representations and information from pre-trained models [22]. TL has attracted a lot of interest and has shown itself to be quite successful in the field of medical imaging. CNN models which are also commonly known as ConvNet that have already been trained on large-scale datasets, models like VGG, ResNet, AlexNet, DenseNet [28] and etc. Used to capture rich visual features that are relevant across different image recognition tasks [24]. These pre-trained models have mastered the ability to recognize and extract both high-level characteristics like object patterns and structures as well as low-level elements like edges, textures, and forms [20].

Transfer learning shows considerable potential for the detection and prediction of various diseases such as Alzheimer's disease. Since it is unusual to have a dataset large enough to train an entire Convolutional Network from scratch (with random initialization), this approach is impractical World Alzheimer Report 2010 [39]. Another option is to use a Conv-Net that has already been trained on a massive dataset like ImageNet [40], as an initialization or a fixed feature extractor for the task at hand. We can use a pre-trained Convolutional Network that has been trained on a sizable dataset like ImageNet [40] rather of creating a Convolutional Network from scratch, which necessitates an impractically big dataset World Alzheimer Report 2010 [39]. Two main transfer learning scenarios typically exist:

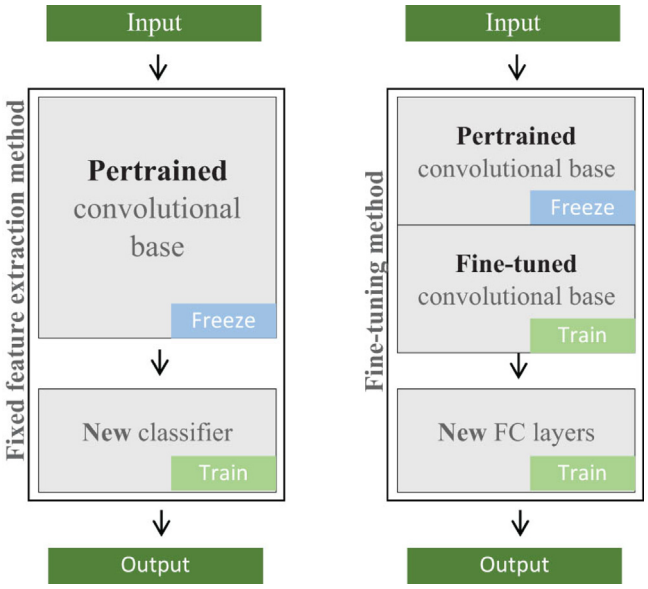

Fig. 1. Transfer learning strategies.

- **ConvNet as a fixed feature extractor**: Using a ConvNet that has been previously trained on ImageNet, we can remove the final fully connected layer and use the remaining structure as a fixed feature extractor for the target dataset. This approach capitalizes on the pre-learned representations from ImageNet, allowing us to extract relevant features without retraining the entire network [41].
- **ConvNet fine-tuning**: A different approach is to replace the classifier's final fully connected layer while also adjusting the parameters of the pretrained network. We are only able to finetune a few higher-level components of the network due to overfitting concerns. This idea is motivated by the observation that earlier features in a ConvNet contain more general features (such color blob detectors or edge detectors) that can be helpful for many tasks [42]. The network's later layers, however, become steadily more attentive on the specifics of the classes in the original dataset. Fig. 1 [43], shows the two different approaches and how they can be used, in the left we have feature extractor approach and on the right fine-tunning approach.

Fig. 2 shows the transformation of the base model to the target model. It illustrates how the pre-trained DenseNet model, which was trained on the dataset ImageNet, was modified. The pretrained model is used as a feature extractor and new layers are added on top for task-specific learning. In this method, only the weights of the newly added layers are trained for the particular task; the weights of the pre-trained model are left fixed during training. This is a typical transfer learning strategy, especially when there is a dearth of labeled data for the target task. Adding a new fully connected layer with three output nodes for the three categories of brain MRI images Mild-Demented (MID), Non-Demented (ND), and Very-Mild-Demented (VMD). This transformation enabled the model to perform the classification task on the new dataset of brain MRI images.

4.1. Significance and exploiting the potential of DenseNet architecture

CNN usage is fairly comparable to that of a standard neural network. A CNN has several layers: an input layer, an output layer, and one or more hidden layers. Convolutional, pooling, or fully connected layers serve as the hidden ones [47]. As the name suggests, ConvNet architectures are built on the explicit premise that the inputs are images, allowing us to hardcode certain characteristics. These then make it